

# Gaze-based Metrics of Cognitive Load in a Conjunctive Visual Memory Task

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## ABSTRACT

Measurement of Cognitive Load (CL) is of considerable importance to Human-Computer Interaction (HCI) as it relates to ease of learn- and usability. Numerous methods have been used for this purpose, both subjective and objective. The former relies on perception of expended effort subject to confounding factors. Among objective measures, eye tracking has demonstrated its effectiveness as a precise and non-invasive alternative. Since eye-tracking indices have mainly been tested in tasks involving single visual properties (e.g., shape), this work aims to compare their sensitivity to conjunctive features. Specifically, this study evaluates the Low/High Index of Pupillary Activity (LHIPA) and microsaccade magnitude in the Color Visual Short-Term Memory (CVSTM) task versus the previously validated n-back task. Results show LHIPA and microsaccade magnitude are as effective in discerning baseline and task in CVSTM as in n-back, replicating earlier results and extending their reliability for evaluation of CL in tasks involving multivalent conjunctive features.

## CCS CONCEPTS

• **Human-centered computing** → **Human computer interaction (HCI)**.

## KEYWORDS

eye tracking, visual attention, cognitive load

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## 1 INTRODUCTION

Cognitive Load (CL) has received considerable attention by the human factors community as it is a measure of working memory demand and information processing linked with task difficulty [Schnotz and Christian 2007]. However, gaze-based CL measurement lacks clarity, with no singular metric standing out from among numerous approaches, e.g., microsaccades, pupil diameter, and blink rate, among others. Moreover, and possibly more importantly, protocols used to validate cognitive load measures often rely on fairly straightforward tasks, e.g., arithmetic, number counting, or n-back, ones that depend on a single feature, not necessarily representative of more realistic scenarios.

The contribution of this paper is validation of the Color Visual Short-Term Memory (CVSTM) task that is based on conjunctive search. We consider this task a step forward in substantiating CL measures, and a lead-in to future studies expanding to conjunctive processing (including visual search) of multivalent features.

## 2 BACKGROUND

Cognitive Load (CL) was first described by Sweller [1988; 1994]. Since then, many methods have been developed for its measurement. Sweller et al. [2011] divided them into indirect measures (i.e., reflecting user performance) through a secondary task, physiological measures, and subjective measures (e.g., questionnaires),

One of the most commonly used subjective measures is the NASA Task Load Index (NASA-TLX) questionnaire, a multi-dimensional rating scale [Hart 2006; Hart and Staveland 1988] extensively employed in experiments of CL [Biondi et al. 2023; Buchner et al. 2021; Emami and Chau 2020]. Minkley et al. [2021] note that subjective measurement of CL has been one of the primary research approaches. However, the technique still lacks validity completeness, both because of a large variety of questionnaires used, and due to various influencing factors such as prior user knowledge, interest, and motivation. For these reasons, subjective measures do not necessarily accurately reflect CL. On the contrary, we contend task complexity can be more precisely measured objectively.

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Objective measures of CL offer a series of advantages over subjective measures. Subjective tools can usually be used only after completion of the task, e.g., with questionnaires such as the NASA-TLX, or other instruments such as think-aloud protocols. However, they tend to lack ecological validity, not to mention unnatural extension of performance (task completion). Among various objective measures, ones measuring performance and psycho-physiological responses offer reduced intrusiveness in task performance and allow measurement and analysis of inter-personal variability. In this work, we are mainly concerned with these types of CL measures, ones that generally do not require physical contact as represented by remote eye-tracking methodology.

Eye movements have been extensively studied for CL measurement, with indices such as blink rate [Chen and Epps 2014b], number and duration of fixations [Fitts et al. 1950; Jacob and Karn 2003; Just and Carpenter 1976], regressions [Azuma et al. 2014], microsaccades [Engbert and Kliegl 2003], saccadic intrusion [Biswas and Prabhakar 2018; Dilli Babu et al. 2019], and pupil diameter measures, e.g., baseline-related task-evoked pupillary response [Ahern and Beatty 1979; Beatty 1982; Beatty and Lucero-Wagoner 2000].

Among these, the latter is popular, since modern eye trackers provide reliable and easy-to-access data that includes pupil diameter [Chen and Epps 2014a; Pflieger et al. 2016; Piquado et al. 2010]. Since its inception by Beatty and Kahneman [1966] as a parametric measure for evaluating working memory load, numerous examples of CL studies exploiting measurement of pupil diameter further tested it in applications involving web maps [Kiefer et al. 2016], arithmetic calculations [Chen and Epps 2014b; Krejtz et al. 2018], decision-making in serious games [Jerčić et al. 2020], n-back [Biondi et al. 2021; Peysakhovich et al. 2017], and simulated driving [Čegovnik et al. 2018; Heeman et al. 2013; Kun et al. 2013; Palinko and Kun 2012] and flight [Peysakhovich et al. 2015; Zheng et al. 2022].

More recently, oscillation of pupil diameter has been used, with the Index of Cognitive Activity (ICA [Marshall 2000, 2002]), the Index of Pupillary Activity (IPA [Duchowski et al. 2018]), and the Low/High Index of Pupillary Activity (LHIPA [Duchowski et al. 2020]). We focus on microsaccades and LHIPA, which, in previous work, have shown reliability to measurement of cognitive load during tasks such as mental counting and n-back [Duchowski et al. 2020]. During some of these prior studies, participants were forced to fix their gaze on a specific point on the display, with sound prompts used to indicate eye movement straying off-target. Although some earlier work did allow for limited movement of gaze, most of these tasks relied on visual stimuli composed of single features, e.g., letters or numbers. The next step in furthering validation of these metrics is to at once replicate earlier results and at the same time extend them with tasks that are more complex in nature, composed of *conjunctive* visual features.

## 2.1 Cognitive Load-Inducing Tasks

The CL literature lacks standardized tasks that can be used to validate CL measurement. One common approach is performing a secondary task [Brünken et al. 2004; Hunter 2021; Park and Brünken 2015; Schoor et al. 2012]. However, this method is mainly used to test the effect of cognitive load on the primary task. Another

technique is creating tasks with different difficulty levels, for example asking participants to do some arithmetic operations such as fraction comparisons [Ikehara and Crosby 2005], addition changing digit numbers [Chen et al. 2011], or addition or subtraction of subsequent numbers [Korbach et al. 2018]. Recently, Deck et al. [2021] described other common CL-inducing tasks, e.g., the memorization of a visual pattern (commonly a series of dots in a grid) followed by a distracting task, where participants must reconstruct the location of the dots [Gerhardt et al. 2016] or identification of a particular position containing one of them [Neys 2006]. A second approach is the n-back task, where participants must indicate if the last element displayed is the same of the  $n^{\text{th}}$  previous one [Appel et al. 2018].

To accomplish the n-back task, participants are required to encode the incoming stimuli, monitor and update the stimulus, and finally compare the currently displayed stimulus with the one presented  $n$  trials earlier. The n-back task is challenging as it requires simultaneous management of multiple cognitive processes [Jaeggi et al. 2010]. The ease of manipulation of the  $n$  factor makes it easy to manage task difficulty, making it a fairly reliable instrument in CL research, with numerous examples [Fridman et al. 2018; He et al. 2019; Ksedžić et al. 2020; Khanam et al. 2022].

Another variation of the n-back task involves the use of time pressure. However, this has been shown to affect mainly social and strategic decisions [Deck and Jahedi 2015; Deck et al. 2021]. Finally, another approach requires memorization of variable-length strings. This variation is based on classic work by Miller [1956] on retention of a maximum number of seven pieces of information plus or minus two. Manipulating the string length can modulate different levels of CL [Allred et al. 2016; Deck and Jahedi 2015].

## 2.2 Color Visual Sort-Term Memory Task

The Color Visual Short-Term Memory (CVSTM) task, derived from work by Luck and Vogel [1997] and Meyerhoff and Gehrer [2017], challenges participants to remember different visual features of a number of objects, e.g., color and position. The short-term retention of visual information is conventionally considered separate from prior semantic knowledge, although recent studies account for long-term memory involvement in everyday visual environments [Xie and Zaghloul 2021]. In the CVSTM task, the number of objects is controlled to vary task difficulty, since many studies have shown that increasing the number of items increases memory load, highlighted by an increase in the magnitude of errors of stimulus recall [Lilburn et al. 2019]. The scope of this task is thus to test short-term memory in terms of encoding and storing information about the stimulus features, using eye-tracking technology [Pavisis et al. 2021].

Feature Integration Theory (FIT) [Treisman and Gelade 1980] describes visual attention to multivalent stimulus features. Developed in the context of visual search, FIT provides insights into how object identification is acquired through integration of stimulus component features. The theory posits *conjunctive search*, in which the objects require integration of their features, distinguishing them from distractors [Humphreys 2016; Singh and Schubert 2021]. FIT has been applied to numerous HCI-related tasks, including image search [Kobayashi et al. 2013], visual saliency computing [Wang et al. 2021], and information visualization [Cai et al. 2015]. The

CVSTM task, although not strictly a visual search task, requires top-down attention to feature binding to encode multivalent stimuli. Indeed, CVSTM requires the use of visual short-term memory for memorization of multiple features of the target (i.e., position and color). To the best of our knowledge, no previous study has examined eye movements during a CL-inducing task based on multivalent features.

The practical implication of this study is the possible application of eye-tracking CL indexes in a visual task (composed of multivalent features) in different HCI tasks with similar characteristics, such as assessing interface quality [Goldberg and Kotval 1999]. For instance, in the practice of interface visual design, human factors guidelines suggest supporting users with feedback, chunking information, and memory aid to reduce experienced memory load [Mazza 2017]. Validating psychophysiological indices in the CVSTM allows reliable studies on visual short-term memory limitations and cognitive load associated with different design approaches based on these guidelines, with the final aim of facilitating creation of easy-to-use and easy-to-learn interfaces.

### 3 EMPIRICAL VALIDATION

Two experiments were run, the n-back and the CVSTM. The first replicates a previous study using 1-back and 2-back tasks, with removal of the fixed-gaze restriction meant to entice the participant to focus at a single point on the screen. The second challenges participants to remember a field of colored dots dispersed over a  $10^\circ \times 8^\circ$  visual field (span).

The experimental design was within-subjects, with difficulty level as independent variable, counterbalanced across participants. Volunteers participated in both tasks, with order counterbalanced as well. We analyzed dependent variables including performance accuracy obtained as the percentage of correct responses in each task. We also analyzed three different gaze metrics: LHIPA, microsaccade magnitude, and rate.

Both experiments used the same experimental setup. The eye tracker was an SR Research EyeLink 1000, sampling at 1000 Hz, with accuracy given by the manufacturer as  $0.25\text{--}0.5^\circ$  visual angle, on average, corroborated by Wang et al. [2017] via root-mean-squared analysis. van der Geest and Frens [2002] reported the EyeLink's precision to be  $0.98^\circ \times 1.05^\circ$ . PsychoPy (version 2022.2.3) [Peirce 2007] was used to write the experimental procedure. Stimuli were presented on a  $2560 \times 1440$  resolution monitor at a viewing distance of 60 cm. A chin rest stabilized head movement. To avoid light interference, the laboratory used for the study possessed no windows. Luminance at the computer screen was 96 Lux, with ambient luminance measured at 550 Lux.

Volunteers ( $N = 24$ ) for the study were recruited verbally and by email (16 M, 8 F with ages in range [19:37] years old,  $M = 24.00$ ,  $SD = 4.94$ ). All were right-handed. Five participants wore vision correction aids (e.g., 2 contact lenses, 3 eyeglasses). Two participants indicated slight color blindness, however, they did not report difficulties in discriminating the CVSTM task colored targets when tested before the experimental trials.

To analyze gaze data, we first pre-processed the pupil diameter signal by removing data 200 ms before the start of, and 200 ms following the end of a blink, as identified by the eye tracker, following

Engbert and Kliegl [2003]. We then computed the indices from the raw signal. Finally, at the end of each task, we collected the subjective experience of participants using the NASA-TLX questionnaire, which allows evaluation of subjective perception of CL using the median of six scales: mental demands, physical demands, temporal demands, performance, effort, and frustration.

#### 3.1 Experiment 1: n-back Task

To compare the sensitivity of the eye-tracking indices to task difficulty, we first used the same experimental task used by Duchowski et al. [2020], without forcing participants to fixate the same spot during the entire experiment. Specifically, this experiment exploits the well-established n-back task, with  $n$  used to evoke different levels of cognitive load.

The experimental protocol was based on Ahern and Beatty [1979] and Appel et al. [2018]. Two types of trials were shown to participants: easy (1-back) and difficult (2-back). Every participant was presented with a series of letters randomly selected from a set  $L = \{C, F, H, S\}$ . The letters were shown one after another and participants had to state if the current letter was the same as the one before (1-back) or as the one that appeared two trials earlier (2-back). Each trial lasts 0.5 seconds with an inter-stimulus duration of 1.5 seconds, with a total trial duration of 2 seconds (see Fig. 1).

Before each block, participants underwent a training phase in which they were presented with 15 trials of the same task they were about to conduct. When they reached an accuracy of at least 60% they started the experiment. After the training block, each participant was presented with a 30 s baseline task, in which they fixated a cross at the center of the screen. Each experimental block consisted of 60 trials, with a total duration of 2 minutes. The letters were displayed at the same position, in the center of the screen. The order of the 1- and 2-back tasks was counterbalanced. At the end of each task, participants responded to the NASA-TLX questionnaire.

#### 3.2 Experiment 2: CVSTM Task

The CVSTM experiment followed the work of Meyerhoff and Gehrer [2017]. The task required remembering the location and color of a series of squares. For each trial, participants were shown a field of colored squares ( $1^\circ \times 1^\circ$ ). In the easy task, 4 squares were displayed, in the difficult task 8. In both tasks, the squares were shown at screen center, within a  $10^\circ \times 8^\circ$  visual angle field of view. Colors were randomly selected from a set consisting of black, white, turquoise, blue, pink, yellow, green, red. Each task was composed of 40 trials. Stimuli were shown for 200 ms. After a short retention interval (900 ms), a probe appeared and participants had to indicate if it consisted of the same features (location and color) as one of the squares shown previously. The probe appeared for 500 ms with inter-stimulus interval of 500 ms. The total duration of each block was 84 s. The procedure is schematically shown in Fig. 2.

Prior to testing, participants completed eight training trials in which the starting set was composed of two elements. When they reached an accuracy of at least 60% they could begin the experiment. After the training block, each experiment was preceded by a 30 s baseline, in which participants fixated a cross at the center of the screen. At the end of each task, participants responded to the NASA-TLX questionnaire.

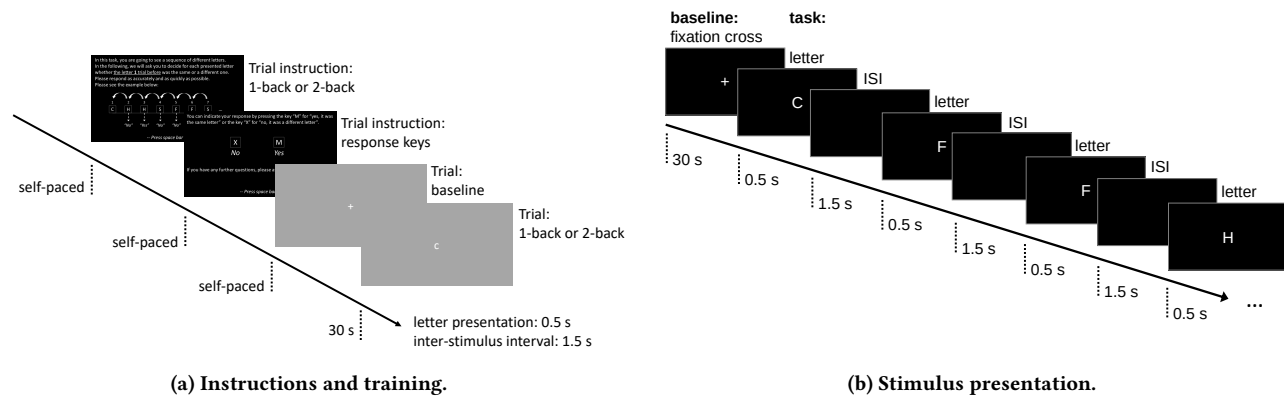


Figure 1: Schematic of the n-back task, including (a) instructions and training preceding (b) stimulus presentation.

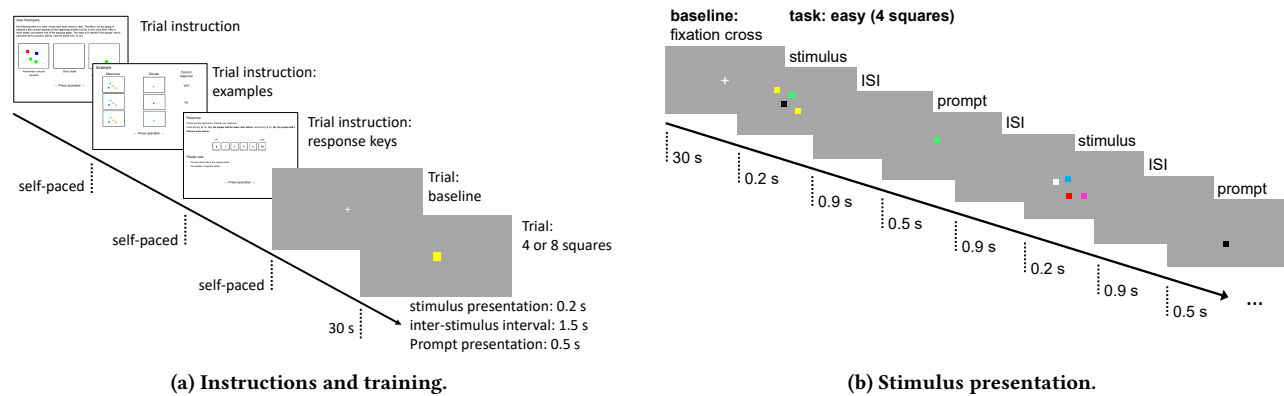


Figure 2: Schematic of the CVSTM task, including (a) instructions and training preceding (b) stimulus presentation.

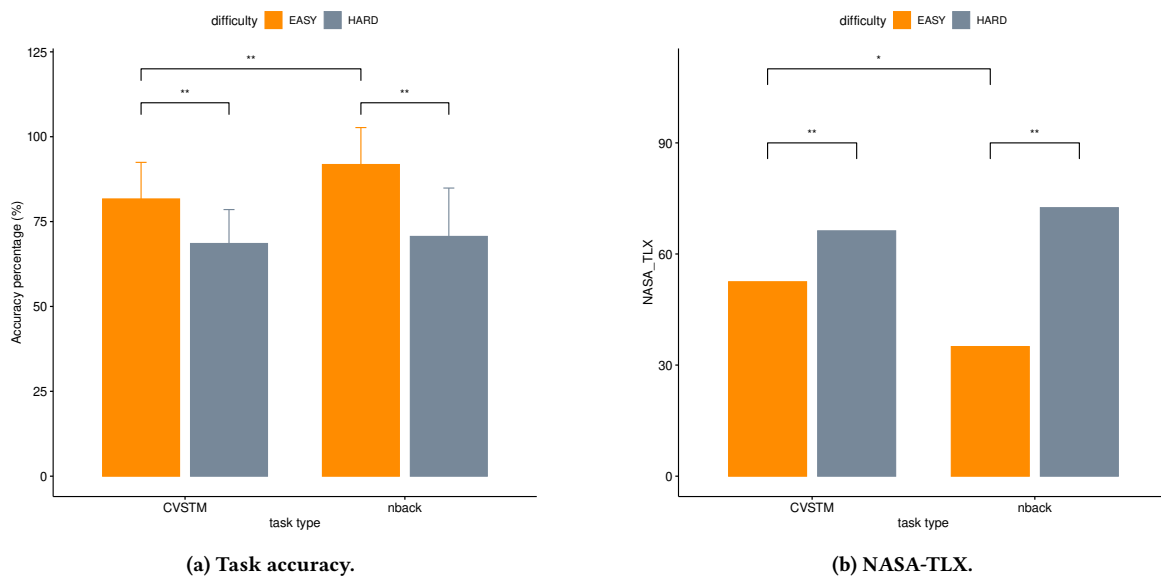
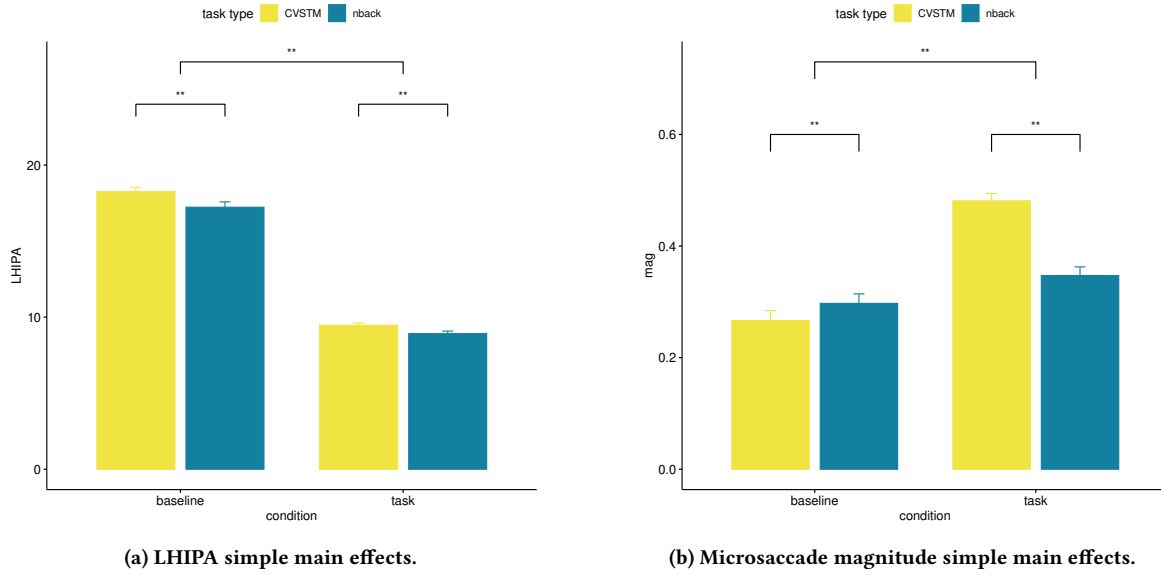


Figure 3: Accuracy and subjective measures, with (a) task accuracy and (b) NASA-TLX median scores. Whiskers represent standard error.



**Figure 4: Results from ANOVA simple main effects, with (a) LHIPA and (b) microsaccade magnitude. Whiskers represent standard error.**

## 4 RESULTS

Analysis used a  $2 \times 2$  within-subjects design, with task difficulty (EASY vs.HARD) and task (CVSTM vs.NBACK) as independent variables. Dependent variables included task accuracy, subjective measures, and gaze-based indices of cognitive processing, e.g., load, including LHIPA and microsaccade magnitude and rate. Gaze-based metrics were contrasted against baseline, serving as two experimental conditions (baseline vs.task).

### 4.1 Accuracy and Subjective Measures

Performance and subjective measures show that participants discriminated between difficulty of each task.

**4.1.1 Accuracy.** A within-subjects two-way repeated-measures ANOVA was run with percentage of correct responses as the dependent variable and task difficulty and type as independent variables. ANOVA showed a statistically significant interaction between task difficulty and task type ( $F(1, 23) = 5.830, p < 0.05, \eta^2 = 0.031$ ), see Fig. 3(a).

A simple main effect of task was significant ( $F(1, 23) = 7.377, p < 0.05, \eta^2 = 0.068$ ). A post-hoc t-test showed the easy task in CVSTM ( $M = 81.7\%, SD = 10.8$ ) yielded significantly fewer correct answers ( $t(23) = -3.13, p < 0.01$ ), than the easy n-back task ( $M = 91.8\%, SD = 10.9$ ). No differences were found between the difficult tasks.

A simple main effect of difficulty was significant ( $F(1, 23) = 106.675, p < 0.01, \eta^2 = 0.363$ ). Post-hoc t-tests showed that the easy CVSTM task ( $M = 81.7\%, SD = 10.8$ ) yielded significantly more correct answers ( $t(23) = 7.07, p < 0.01$ ) than the difficult task ( $M = 68.5\%, SD = 10$ ). The t-tests also revealed that the easy n-back task ( $M = 91.8\%, SD = 10.9$ ) yielded significantly more correct answers ( $t(23) = 7.67, p < 0.01$ ) than the difficult task ( $M = 70.6\%, SD = 14.3$ ).

**4.1.2 NASA-TLX.** NASA-TLX results were analyzed using paired Wilcoxon tests with Benjamini-Hochberg correction, see Fig. 3(b). Results showed that the CVSTM easy task ( $Med = 52.5$ ) was considered significantly easier ( $V = 18, p < 0.01$ ) than the difficult task ( $Med = 66.2$ ). A similar result was found in the n-back task, where the easy task ( $Med = 35$ ) was considered significantly easier ( $V = 0, p < 0.01$ ) than the difficult task ( $Med = 72.5$ ). Results also showed that the CVSTM easy task was considered significantly harder than the easy n-back task ( $V = 217.5, p < 0.05$ ).

### 4.2 Gaze Measures

Simple main effects of LHIPA and microsaccade magnitude show their ability to discriminate between task and baseline conditions, with LHIPA decreasing and microsaccade magnitude increasing during the task execution.

**4.2.1 LHIPA.** We ran a  $2 \times 2 \times 2$  repeated measures three way ANOVA with task, condition, and task difficulty as within-subjects independent factors. Results are shown in Fig. 4(a) with LHIPA as the dependent measure. The test did not show a significant three-way interaction. ANOVA revealed a simple two-way interaction between condition and task ( $F(1, 23) = 7.914, p < 0.01, \eta^2 = 0.005$ ). The simple main effect of condition was significant ( $F(1, 23) = 700.246, p < 0.01, \eta^2 = 0.857$ ). The simple main effect of task was significant ( $F(1, 23) = 23.521, p < 0.01, \eta^2 = 0.048$ ). The simple main effect of task difficulty was not significant.

Post-hoc pairwise t-tests, with task as independent variable, revealed that LHIPA at CVSTM baseline ( $M = 18.2, SD = 1.77$ ) was significantly greater ( $t(23) = 4.45, p < 0.01$ ), than at n-back baseline ( $M = 17.2, SD = 2.30$ ). Results also revealed that LHIPA in the CVSTM task ( $M = 9.5, SD = 1.02$ ) was significantly greater ( $t(23) = 4.68, p = 0.01$ ), than at the n-back task ( $M = 8.92, SD = 1.08$ ).

Pairwise t-tests with condition as independent variable showed that CVSTM baseline LHIPA was significantly greater ( $t(23) = 27.6, p < 0.01$ ), than the task LHIPA. A similar result was found for the n-back task ( $t(23) = 23.9, p < 0.01$ ). A simple main effect was significant for condition and task. Post-hoc pairwise t-test comparisons confirmed significant differences in LHIPA between CVSTM and n-back ( $t(23) = 5.63, p < 0.01$ ), and between baseline and task ( $t(23) = 44.5, p < 0.01$ ).

**4.2.2 Microsaccade Magnitude and Rate.** We ran a  $2 \times 2 \times 2$  repeated measures three way ANOVA with task, condition, and task difficulty as within-subjects independent factors. Microsaccade magnitude was used as the dependent measure. ANOVA did not show a significant three-way interaction.

A simple main effect was significant for condition ( $F(1, 23) = 92.014, p < 0.01, \eta^2 = 0.268$ ) and task ( $F(1, 23) = 38.561, p < 0.01, \eta^2 = 0.053$ ). Post-hoc pairwise comparisons confirmed the differences in microsaccade magnitude between baseline and task ( $t(23) = -10.6, p < 0.01$ ), and between CVSTM and n-back ( $t(23) = 3.21, p < 0.01$ ). Results are shown in Fig. 4(b).

ANOVA revealed a simple two-way interaction between condition and task ( $F(1, 23) = 95.7, p < 0.01, \eta^2 = 0.124$ ). A simple main effect of task was significant ( $F(1, 23) = 38.561, p < 0.01, \eta^2 = 0.053$ ). The simple main effect of condition was significant ( $F(1, 23) = 92.014, p < 0.01, \eta^2 = 0.268$ ). Post-hoc pairwise t-tests, with condition as independent variable, revealed that microsaccade magnitude during CVSTM baseline ( $M = 0.266, SD = 0.124$ ) was significantly lower ( $t(23) = -12.8, p < 0.01$ ) than during CVSTM task ( $M = 0.481, SD = 0.092$ ). Similar results were found for n-back baseline ( $M = 0.297, SD = 0.118$ ), and ( $t(23) = -3.20, p < 0.01$ ) n-back task ( $M = 0.347, SD = 0.108$ ).

Pairwise t-tests, with task as independent variable showed that, in the baseline condition, microsaccade magnitude of CVSTM was significantly lower ( $t(23) = -2.88, p < 0.01$ ) than n-back. In the task condition, microsaccade magnitude of CVSTM was significantly greater ( $t(23) = 10.4, p < 0.01$ ) than n-back.

ANOVA revealed a simple two-way interaction between condition and task difficulty ( $F(1, 23) = 7.038, p < 0.05, \eta^2 = 0.006$ ). The simple main effect of task difficulty was not significant. The simple main effect of condition was significant ( $F(1, 23) = 92.014, p < 0.01, \eta^2 = 0.297$ ).

Post-hoc pairwise t-tests, with condition as independent variable, revealed that, in the easy task, microsaccade magnitude during baseline ( $M = 0.287, SD = 0.122$ ) was significantly lower ( $t(23) = -6.82, p < 0.01$ ) than during the tasks ( $M = 0.402, SD = 0.086$ ). Similar results ( $t(23) = -11.2, p < 0.01$ ) were obtained between hard task baseline ( $M = 0.277, SD = 0.107$ ) and task ( $M = 0.427, SD = 0.098$ ).

A  $2 \times 2 \times 2$  repeated-measures three-way ANOVA with task, condition, and task difficulty as within-subjects independent factors did not show any significant interaction, given microsaccade rate as the dependent measure.

## 5 DISCUSSION & CONCLUSION

Accuracy measures suggest that the multivalent features of the easy CVSTM task indicate it as significantly harder than the monovalent visual nature of the easy n-back, agreeing with subjective

impressions of significantly increased difficulty of the former versus the latter. Gaze data shows that LHIPA discriminates between baseline and task conditions (see Fig. 4(a)), confirming previous results [Duchowski et al. 2020]. As before, LHIPA did not indicate differences between task difficulties within each task. However, LHIPA appears to discriminate between the CVSTM and n-back, indicating greater CL for the n-back task, which is incongruent with both performance and subjective measures.

Results highlight a fundamental difference between the two main eye movement indices, with LHIPA indicating n-back as the harder task, and the reverse for microsaccade magnitude. One possible explanation is that LHIPA is more sensitive to tasks involving short-term memory while microsaccades are more sensitive to *visual* short-term memory.

Another explanation for microsaccade magnitude response may take into account the nature of the two tasks. The n-back requires fixating a single spot on the screen. Once the stimulus appears, one need no longer fixate it, allowing vision to fade while concentrating on recalling the previous stimuli, thus reducing the need for microsaccades to maintain focus. Despite microsaccades being largely involuntary, they can be reduced by voluntary and stable fixation on target [Martinez-Conde et al. 2013], possibly explaining gaze behaviour in the n-back task, where visual control is not linked to image preservation, one of the reasons for microsaccades [Martinez-Conde et al. 2006]. In the CVSTM task, participants need to maintain visual contact with the stimulus to retrieve all stimulus features for the entirety of its duration. In this case, microsaccade magnitude could increase to prevent image fading. As reported by Martinez-Conde et al. [2006], during transition from fading to visibility, microsaccade magnitude increases, suggesting a relationship between their production and visibility during fixations.

The CVSTM appears to elicit greater task difficulty than n-back, as reflected by performance, subjective measures, and microsaccade magnitude. It is possible to conclude that CVSTM can be a viable tool to evoke mental effort in scenarios approximating real-world human-computer interaction in conjunction with the use of eye-tracking metrics. The empirically tested indices herei can provide reliable methods to potentially rate interface quality, user engagement, or satisfaction [Schmutz et al. 2009]. Moreover, measuring CL with theses measures could provide more objective, convenient, detailed, and less invasive alternatives to self-reported metrics. Finally, another HCI domain which could benefit from reliable visual short-term memory-related CL measures is adaptive interfaces. These are relatively new systems which modulate their activity according to user cognitive availability [Mingardi et al. 2020], requiring collection of accurate data and CL measures. Present findings, subject to future replication, indicate that the CVSTM is a good starting point for future research on cognitive load with multivalent, conjunctive visual features.

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